Course Title	**<u>Mathematical Foundation for Data Science</u>** Course Instructor: Prof. Bhaskar Chaudhury		
Course Code	DS601	Credit Structure	3-1-0-4
Course Format	Three lecture sessions and One Tutorial Session per week		
Category	Core	Semester	Autumn Semester (AY 2026-27)
Program	PG - M.Sc(Data Science)		
Course Objectives/ Brief Course Description	To provide a strong foundation of the important mathematical concepts for modern data sciences, to be covered in three primary modules: M1) Probability and Random Variables and M2) Linear Algebra and M3) Calculus. To empower students with the mathematical insight needed to move beyond using data science tools to understanding, designing, and improving data driven and ML algorithms.		
Evaluation/ Grading Policy	First Insem (30%), Second Insem (30%) and End-Sem (40%) Grading scheme is relative and depends on both: class performance and minimum expectation from a student.		
Course Materials/ References	1. Gilbert Strang, Linear Algebra and its applications. 2. Linear Algebra and Learning from Data by Gilbert Strang 3. A First course in Probability by Sheldon Ross. 4. Mathematics for Machine Learning, Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong, Cambridge University Press 5. Coding the Matrix: Linear Algebra through Applications to Computer Science, by Philip Klein.		
Detailed Course Content	**APPENDIX - I**		

Course Outcome:

CO1: Apply the mathematical foundations of probability, linear algebra, and calculus to solve problems in data science.
CO2: Analyze the mathematical principles underlying machine learning algorithms and interpret their behavior with greater insight.
CO3: Formulate and solve data science problems using appropriate mathematical models. Course forms the foundation of most methods used in machine learning.

Program Outcomes

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X		X								X

APPENDIX- I: Detailed Course Content (Session-wise/ Module-wise)

Module	Title & Description	No of Lectures
M1A: Basics of Probability	Probability definition, Sample space, events, principles of counting, sampling table, Axioms of Probability, Inclusion–exclusion principle, Independence, Conditional Probability, Bayes theorem (Prior and Posterior; probabilities of hypotheses when given evidence), Law of total Probability, Simpson's paradox.	**6**
M1B: Random Variables, Probability Distribution	Random variables (RV) - Bernouli, Discrete uniform random variable, Binomial RV, Geometric RV, Discrete and Continuous RVs, Probability Mass Function, Binomial Distribution, Probability Density functions Expected value of a RV, Expected value of a function of a random variable, Linearity of expectation, Variance and standard deviation, Cumulative Distribution Function, Multiple Random Variables and Joint PMFs, Normal RV/ Gaussian RV (standard Normal) - some problems using "Areas under the standard normal curve" table, General Normal RV Additional topics (optional): Exponential Random Variables and PDF, Memorylessness of the exponential distribution, Poisson Distribution.	**10**
M2A: Vector Spaces and Subspaces, Matrix	Data as Vectors, Operations on Vectors. Linear Dependence and Independence. Spaces of Vectors, subspaces, Span, Basis and Dimension, Vectors and Linear Combinations, Norms, Representation of linear transformations as a matrix. Four fundamental subspaces associated with a matrix, Dimensions of the Four Subspaces, transpose, rank, determinant, matrix inverse, Determinants (important properties of determinants, Det of a triangular matrix, finding determinant of big matrix using product of pivots), trace. Special matrices: square matrix, identity matrix, triangular matrix, idea about sparse and dense matrix, symmetric matrix.	**6**
M2B: Solving Linear Equations	Vectors and Linear Equations, The Idea of Elimination, Elimination Using Matrices, Rules for Matrix Operations, Inverse Matrices, Elimination, Gauss elimination, Gauss-Jordan elimination, Row reduced form. Row reduced echelon form. Factorization: A = LU, Transposes and Permutations, The Nullspace of A: Solving Ax = 0 and Rx = 0, The Complete Solution to Ax = b.	**6**
M2C: Orthogonality and Matrix Diagonalization	Orthogonality of the Four Subspaces, Projections, Least Squares Approximations, Orthonormal Bases and Gram-Schmidt. Eigenvectors and Eigenvalues, Diagonalizing a Matrix, SVD.	**5**
M3: Calculus	Functions, continuous functions, differentiable functions, function of several variables, gradient of a function, and directional derivatives. Minima of a convex function, gradient descent method.	**5**
	Case Studies using Real Data	**2**

Learn the mathematics behind data science and AI/ML - not just to use intelligent algorithms, but to understand, improve, and create them.
